



Brief paper

A trajectory tracking control law for a quadrotor with slung load[☆]David Cabecinhas^{a,b,*}, Rita Cunha^{c,b}, Carlos Silvestre^{a,1}^a Department of Electrical and Computer Engineering, Faculty of Science and Technology of the University of Macau, Macau, China^b Institute for Systems and Robotics, Instituto Superior Técnico, Universidade de Lisboa, Portugal^c Department of Electrical Engineering and Computer Engineering, Instituto Superior Técnico, Universidade de Lisboa, Portugal

ARTICLE INFO

Article history:

Received 22 January 2018

Received in revised form 22 November 2018

Accepted 28 March 2019

Available online 24 May 2019

Keywords:

Slung load control

Quadrotor vehicle

Lyapunov methods

Backstepping

ABSTRACT

We present a trajectory tracking controller for the full dynamics of a quadrotor vehicle carrying a slung load attached by a string. The full dynamic system is modeled as two connected subsystems, the string-load subsystem, with dynamics identical to that of a standard quadrotor in free flight, and the quadrotor subsystem with attitude kinematics and dynamics. A trajectory tracking controller for the position of the point-mass load is designed based on existing Lyapunov-based trajectory tracking controllers for free flying quadrotors which are further backstepped through the quadrotor attitude dynamics. A parameterized Lyapunov function is provided for the full system dynamics with a negative semi-definite time derivative. The proposed controller is proven to drive the load position error to zero and the origin of the error system is exponentially stable. Simulation results attest the performance of the proposed controller for aggressive trajectories and its robustness and validity are further highlighted by experimental results with a model-scale vehicle and slung load.

© 2019 Elsevier Ltd. All rights reserved.

1. Introduction

The last decade oversaw the first steps and boom of research of slung load transportation with model-scale autonomous rotorcraft. Work on cranes and full size helicopters carrying slung loads provided the inspiration for the original works using model-scale rotorcraft (Bernard & Kondak, 2009). These first attempts were based on inner–outer loop designs and linearization of the vehicle, string and load systems. The load was modeled as a mass point and the control solution involved estimation of string tension forces, based on a flexible string dynamic model obtained using computer software. A comprehensive dynamic model that allows for slack strings and aerodynamic drag on the load was developed in Bisgaard, Bendtsen, and Cour-Harbo (2009).

The use of multiple small-scale aerial vehicles for transport of a slung rigid body was pioneered in Michael, Fink, and Kumar (2011) and the problem from a robotics inspired perspective

where the inputs are the quadrotor positions and the system acts effectively as a parallel manipulator.

Other early works (Faust, Palunko, Cruz, Fierro, & Tapia, 2013; Palunko, Cruz, & Fierro, 2012) focused on trajectory generation for a slung load system resulting in optimal swing-free trajectories. The resulting trajectory for the quadrotor vehicle was then tracked by the vehicle but no feedback from the load position or string angle was used to improve control performance and the control derivation relies on linearization.

More recently, approaches rooted in geometric control, starting with Sreenath, Lee and Kumar (2013b) and Sreenath, Michael and Kumar (2013a), have developed the complete system dynamics and controllers directly on the nonlinear configuration manifold in a coordinate-free form. Moreover, these works consider explicitly the coupling between load and quadrotor dynamics. The resulting controllers are thus almost-global and the load is able to undergo large swings while under feedback control. While Sreenath, Michael et al. (2013a) restricted itself to the planar case, experimental results for the geometric controller (Sreenath, Lee et al., 2013b) were later presented in Tang and Kumar (2015), which focuses on trajectory generation while relaxing the condition that the string be always taut.

Typically, coordinate-free methods can be applied to obtain almost-global controller for a vehicle transporting a payload whereas when dealing with flexible strings, due to the infinite dimensionality of the problem, one cannot escape the use of approximations and linearization methods. It is also common to simplify the control problem by considering that the inner-loop

[☆] The material in this paper was not presented at any conference. This paper was recommended for publication in revised form by Associate Editor Peng Shi under the direction of Editor Thomas Parisini.

* Corresponding author at: Department of Electrical and Computer Engineering, Faculty of Science and Technology of the University of Macau, Macau, China.

E-mail addresses: dcabecinhas@um.edu.mo (D. Cabecinhas), rita@isr.tecnico.ulisboa.pt (R. Cunha), csilvestre@um.edu.mo (C. Silvestre).

¹ On leave from the Instituto Superior Técnico, Universidade de Lisboa, Portugal.

Table 1
Symbols in the quadrotor–string–load system.

$m_Q, m_L \in \mathbb{R}$	Mass of quad, load
$J \in \mathbb{R}^{3 \times 3}$	Quad inertia
$\ell, g \in \mathbb{R}$	String length and gravitational const.
$\mathbf{p}_Q, \mathbf{p}_L \in \mathbb{R}^3$	Position of the quad, load, in $\{I\}$
$\mathbf{v}_Q, \mathbf{v}_L \in \mathbb{R}^3$	Velocity of the quad, load, in $\{I\}$
$\mathbf{r}_L \in \mathbb{S}^2$	Unit vector from quad to load in $\{I\}$
$\Omega_L \in \mathbb{R}^3$	String angular velocity in $\{I\}$
$T_L \in \mathbb{R}, \tau_L \in \mathbb{R}^3$	String tension and ang. accel. in $\{I\}$
$R_Q \in \text{SO}(3)$	Rotation matrix from $\{B\}$ to $\{I\}$
$\omega_Q \in \mathbb{R}^3$	Quad angular velocity in $\{B\}$
$T_Q \in \mathbb{R}, \mathbf{n}_Q \in \mathbb{R}^3$	Quad thrust force and torque in $\{B\}$
$\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3 \in \mathbb{R}^3$	Unit vectors along the reference frame

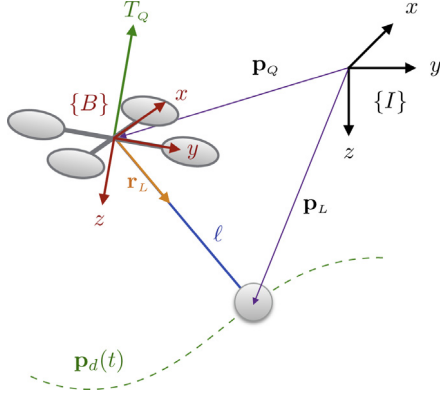


Fig. 1. Quadrotor and slung load system.

corresponding to the quadrotor orientation is attained instantly, or at least in finite time, thereby suppressing the coupling of the quadrotor dynamics on the string and load subsystems.

In this brief we propose a novel look at the slung load system. By drawing a parallel between the string–load subsystem, driven by a fully-actuated vehicle, and a standard quadrotor model in free flight, existing position tracking controllers for quadrotors can be adapted to slung load position tracking. Following the proposed control design, a Lyapunov function can be constructed for the overall closed-loop system resulting in a position error system proven to be almost-globally asymptotically stable and locally exponentially stable. Since no conservatism is introduced, the proposed method has potentially better performance and convergence regions than perturbation theory or inner–outer loop based controllers.

2. Dynamic model

The quadrotor, string, and slung load system is composed of a quadrotor vehicle, a point mass load, and a massless string of fixed length ℓ connecting the quadrotor to the load, as depicted in Fig. 1. The variables used are summarized in Table 1. The skew-symmetric matrix $S(\mathbf{a})$ encodes the cross-product by $\mathbf{a}$ resulting in $S(\mathbf{a})\mathbf{b} := \mathbf{a} \times \mathbf{b}$ for any vectors $\{\mathbf{a}, \mathbf{b}\} \in \mathbb{R}^3$. The projection onto the plane orthogonal to a unit vector is $\Pi_{\mathbf{a}} := -S(\mathbf{a})^2$.

The coordinate-free dynamics for the complete system are

$$\dot{\mathbf{p}}_L = \mathbf{v}_L, \quad (1a)$$

$$\dot{\mathbf{v}}_L = -\frac{1}{m_Q} T_L \mathbf{r}_L + g \mathbf{e}_3, \quad (1b)$$

$$\dot{\mathbf{r}}_L = S(\Omega_L) \mathbf{r}_L, \quad (1c)$$

$$\dot{\Omega}_L = \Pi_{\mathbf{r}_L} \tau_L \quad (1d)$$

$$\dot{R}_Q = R_Q S(\omega_Q), \quad (1e)$$

$$\dot{\omega}_Q = \tau_Q, \quad (1f)$$

where

$$T_L = \frac{m_L}{m_Q + m_L} \mathbf{r}_L^T T_Q \mathbf{r}_Q + \frac{m_L m_Q \ell}{m_Q + m_L} \|\omega_L\|^2, \quad (2)$$

$$\tau_L = -\frac{1}{m_Q \ell} S(\mathbf{r}_L) T_Q \mathbf{r}_Q, \quad (3)$$

$$\tau_Q = J^{-1}(\mathbf{n}_Q - S(\omega)J\omega). \quad (4)$$

These can be derived from first principles by solving Newton's law equations for each rigid body, together with the connection constraint

$$\mathbf{p}_L = \mathbf{p}_Q + \mathbf{r}_L \ell, \quad (5)$$

or by application of the extended Hamilton's principle with variations on the configuration manifold (Sreenath, Lee & Kumar, 2013a). The state vector is $(\mathbf{p}_L, \mathbf{v}_L, \mathbf{r}_L, \Omega_L, R_Q, \omega_Q)$ and the input vector is the quadrotor actuation $(T_Q, \mathbf{n}_Q)$. For simplicity, in the remainder of the paper we consider τ_Q as input since, inverting (4), one can always find the appropriate $\mathbf{n}_Q$ that generates any τ_Q . The dynamic system can be thought as being comprised of three coupled subsystems, where (1a)–(1b) describe the load subsystem, (1c)–(1d) the string subsystem and finally (1e)–(1f) correspond to the quadrotor attitude kinematics and dynamics. The quadrotor position and velocity can be derived from the load position using the relation (5) and its time derivative.

The complete system dynamics (1) are differentially flat (Sreenath & Kumar, 2013; Sreenath, Michael & Kumar, 2013b) with flat output given by the load position $\mathbf{p}_L$ and the quadrotor yaw angle. This property indicates that the feasible trajectories are completely characterized by the flat output and its derivatives and thus the system is linearizable by dynamic feedback into a chain of integrators of the flat output. Flatness also indicates that a modified backstepping procedure can be used to design a tracking controller, considering a sufficiently smooth trajectory for the flat output as reference to be tracked. This property parallels the free flight quadrotor since the dynamic system is not in strict feedback form but, due to differentially flatness, designing backstepping controllers is still possible. The dynamics for a free flight quadrotor system are

$$\dot{\mathbf{p}} = \mathbf{v},$$

$$\dot{\mathbf{v}} = -\frac{1}{m} T \mathbf{r}_3 + g \mathbf{e}_3,$$

$$\dot{R} = R S(\omega),$$

$$\dot{\omega} = \tau.$$

Factoring out the yaw movement and introducing the change of variables

$$\Omega = R \Pi_{\mathbf{e}_3} \omega,$$

$$\tau' = R \Pi_{\mathbf{e}_3} \tau,$$

result in the reduced system dynamics

$$\dot{\mathbf{p}} = \mathbf{v}, \quad (6a)$$

$$\dot{\mathbf{v}} = -\frac{1}{m} T \mathbf{r}_3 + g \mathbf{e}_3, \quad (6b)$$

$$\dot{\mathbf{r}}_3 = S(\Omega) \mathbf{r}_3, \quad (6c)$$

$$\dot{\Omega} = \Pi_{\mathbf{r}_3} \tau', \quad (6d)$$

where $\mathbf{r}_3 = \mathbf{Re}_3$. The parallel between the load-string (1a)–(1d) and the free flight quadrotor dynamics (6a)–(6d) is now immediately clear. In (1a)–(1d), the string tension T_L takes the role of the thrust actuation T and τ_L takes the role of the modified angular acceleration τ' . Using a more compact notation and evidencing the affine nature of the actuation, the quadrotor model with state $\mathbf{x} = (\mathbf{p}, \mathbf{v}, \mathbf{r}_3, \Omega)$ and inputs T and τ' can be written as

$$\dot{\mathbf{x}} = f(\mathbf{x}) + G(\mathbf{x}) \begin{bmatrix} T \\ \tau' \end{bmatrix}, \quad (7)$$

with $f(\mathbf{x}) : \mathbb{R}^{12} \rightarrow \mathbb{R}^{12}$ and $G(\mathbf{x}) : \mathbb{R}^3 \rightarrow \mathbb{R}^{12}$ defined according to (6). Noting that T_L , τ_L are affine functions of T_Q , τ_Q , which according to (2) and (3) can be written as

$$\begin{bmatrix} T_L \\ \tau_L \end{bmatrix} = H(\mathbf{x}_L) T_Q \mathbf{r}_Q + h(\mathbf{x}_L), \quad (8)$$

it follows that the quadrotor-load model with states $\mathbf{x}_L = (\mathbf{p}_L, \mathbf{v}_L, \mathbf{r}_L, \Omega_L)$, R_Q , ω_Q and inputs T_Q , τ_Q can be written compactly as

$$\dot{\mathbf{x}}_L = f_L(\mathbf{x}_L) + G_L(\mathbf{x}_L) T_Q^* \mathbf{r}_Q^* + G_L(\mathbf{x}_L) (T_Q \mathbf{r}_Q - T_Q^* \mathbf{r}_Q^*) \quad (9a)$$

$$\dot{R}_Q = R_Q S(\omega_Q) \quad (9b)$$

$$\dot{\omega}_Q = \tau_Q. \quad (9c)$$

where $T_Q^* \mathbf{r}_Q^*$ are *virtual inputs* for the string-load subsystem and $f_L(\mathbf{x}_L) = f(\mathbf{x}_L) + G(\mathbf{x}_L) h(\mathbf{x}_L)$ and $G_L(\mathbf{x}_L) = G(\mathbf{x}_L) H(\mathbf{x}_L)$. The virtual inputs T_Q^* and $\mathbf{r}_Q^*$ can be seen as the desired stabilizing inputs for the string-load subsystem if the quadrotor attitude dynamics are neglected. The first two terms of (9a) parallel then the free flight quadrotor system (7) and the latter term in (9a) can then be regarded as a *quadrotor actuation error* term.

3. Controller design

In light of the similarities between the string-load subsystems and a free flying quadrotor, it is natural to use the existing body of work on quadrotor free flight control as a starting point. Any existing trajectory tracking controller for a free flying quadrotor for which a Lyapunov function and time derivative are known, e.g. Cabecinhas, Cunha, and Silvestre (2015) and Hua, Hamel, Morin, and Samson (2009), can be used with the proposed approach. The actuation error between the ideal tracking actuation and the current quadrotor actuation, $T_Q \mathbf{r}_Q - T_Q^* \mathbf{r}_Q^*$, is then backstepped through the (9b)–(9c) subsystem to obtain the final control inputs T_Q and τ_Q . Let

$$\mathbf{z} = \zeta(\mathbf{x}, \mathbf{p}_d, \dots, \mathbf{p}_d^{(3)})$$

denote the error state for the tracking of the trajectory $\mathbf{p}_d(t)$ by a free flying quadrotor vehicle. The exact definition of the error $\mathbf{z}$ will depend naturally on the free flight controller chosen to bootstrap the slung-load controller. Furthermore, let $V(\mathbf{z})$ be a Lyapunov function for a trajectory tracking free flying quadrotor dynamic system and let $T^*(\mathbf{z})$ and $\tau^*(\mathbf{z})$ denote the closed-loop inputs that ensure trajectory tracking and render $\dot{V}(\mathbf{z}) \leq -W(\mathbf{z})$ with $W(\mathbf{z})$ a positive semi-definite function. Introducing the error notation

$$\mathbf{z}_L = \zeta(\mathbf{x}_L, \mathbf{p}_d, \dots, \mathbf{p}_d^{(3)}), \quad (10)$$

the vehicle's *virtual* inputs for the string-load system can be computed by inverting (8) as

$$\begin{aligned} T_Q^* \mathbf{r}_Q^* = & \left(\frac{m_L + m_Q}{m_L} T_L^*(\mathbf{z}_L, \mathbf{p}_d^{(4)}) - m_Q \ell \|\omega_L\|^2 \right) \mathbf{r}_L \\ & + m_Q \ell S(\mathbf{r}_L) \tau_L^*(\mathbf{z}_L, \mathbf{p}_d^{(4)}) \end{aligned} \quad (11)$$

The time derivative $\dot{V}(\mathbf{z}_L)$, factoring out the under-actuation error, can be written as

$$\dot{V}(\mathbf{z}_L) = -W(\mathbf{z}_L) + \frac{\partial V}{\partial \mathbf{x}_L} G_L(\mathbf{x}_L) (\mathbf{r}_Q T_Q - \mathbf{r}_Q^* T_Q^*).$$

The backstepping procedure can now be iterated as if for a normal underactuated quadrotor. Following Cabecinhas et al. (2015), the thrust is used to cancel the component of $\mathbf{r}_Q^* T_Q^*$ along $\mathbf{r}_Q$ and the angular error between $\mathbf{r}_Q$ and $\mathbf{r}_Q^*$ is backstepped through the quadrotor attitude subsystem. Setting the thrust actuation

$$T_Q = \mathbf{r}_Q^T \mathbf{r}_Q^* T_Q^*$$

leads to the closed-loop Lyapunov time derivative

$$\dot{V}(\mathbf{z}_L) = -W(\mathbf{z}_L) + \frac{\partial V}{\partial \mathbf{x}_L} G_L(\mathbf{x}_L) T_Q^* R_Q \Pi_{\mathbf{e}_3} R_Q^T \mathbf{z}_Q$$

where $\mathbf{z}_Q$ denotes the quadrotor attitude error given by

$$\mathbf{z}_Q = \mathbf{r}_Q - \mathbf{r}_Q^* \quad (12)$$

and a new Lyapunov function is constructed as

$$V_2 = V + L_2 \frac{1}{2} \mathbf{z}_Q^T \mathbf{z}_Q.$$

where $L_2 > 0$ is an arbitrary matching parameter. The new Lyapunov time derivative features explicitly the quadrotor angular velocity actuation

$$\dot{V}_2 = -W_2(\mathbf{z}_L, \mathbf{z}_Q) - L_2 \mathbf{z}_Q^T R_Q S(\mathbf{e}_3) (\omega_Q - \omega_Q^*)$$

with

$$W_2(\mathbf{z}_L, \mathbf{z}_Q) = W(\mathbf{z}_L) + k_Q \mathbf{z}_Q^T \Pi_{\mathbf{r}_Q} \mathbf{z}_Q,$$

$$\begin{aligned} \omega_Q^* = & S(\mathbf{e}_3) R_Q^T \left(S(\mathbf{z}_Q) \mathbf{r}_Q^* - \frac{k_Q}{L_2} \mathbf{z}_Q \right. \\ & \left. + \frac{1}{L_2} G_L^T(\mathbf{x}_L) \frac{\partial V}{\partial \mathbf{x}_L} T_Q^* \right). \end{aligned}$$

Given that the final integrator chain (9c) is fully actuated, a backstepping iteration with the angular velocity error

$$\mathbf{z}_\omega = \Pi_{\mathbf{e}_3} (\omega_Q - \omega_Q^*) \quad (13)$$

results in a final Lyapunov derivative that can be rendered negative semi-definite with the appropriate control actuation τ_Q . The degree of freedom in the third component of τ_Q , due to the projection in (13), can be independently exploited to control the quadrotor yaw angle.

The quadrotor and slung load dynamics (1) are valid only under the condition that the string connecting the load to the quadrotor is under tension. The following Assumptions capture the taut string condition and ensure $\mathbf{r}_Q^*$ is well-defined.

Assumption 1. Throughout the trajectory tracking maneuver the string connecting the load to the quadrotor is always under tension, corresponding to $T_L > 0$ for all time, with T_L as in (2).

Assumption 2. Throughout the trajectory tracking maneuver the condition $\mathbf{r}_Q^* T_Q^* \neq \mathbf{0}$ is always verified.

Adequate trajectory planning and generation is needed to avoid trajectories that make the quadrotor descent too quickly and possibly violate Assumption 2. In practice, this exceptional situation only occurs for high descent acceleration trajectories, which are not typical of load transportation. The quadrotor and slung load system is differentially flat and, given a load trajectory $\mathbf{p}_L$ and its time derivatives, it is straightforward to plug them

in (1) and compute the evolution of the system states and inputs that correspond to that specific trajectory. This property can be exploited to construct nominal reference trajectories that maintain a nominal minimum string tension and a minimum thrust. In addition, these safeguard distances also provide a metric for the initial error that the closed-loop system tolerates such that singularities or model breaches are avoided. The following Theorem encodes and consolidates the main stability and tracking properties of the proposed controller.

Theorem 3. Consider the dynamic system (1), representing the dynamics of a load being slung from a quadrotor vehicle using a taut string, and a desired trajectory for the load $\mathbf{p}_d(t)$ of class at least C^6 . Let $T^*(\mathbf{z}_L)$ and $\tau^*(\mathbf{z}_L)$ be closed-loop actuations for a quadrotor vehicle, and $\mathbf{z}_L$ its error state, for which a Lyapunov function $V(\mathbf{z}_L)$ and its time derivative are known, such that the resulting error system has an almost globally asymptotically stable equilibrium point at the origin, locally exponentially stable, and $\dot{V}(\mathbf{z}_L) \leq -W(\mathbf{z}_L)$ with $W(\mathbf{z}_L)$ a positive semi-definite function. Then, under Assumptions 1 and 2, for the closed-loop system resulting from the use of the inputs

$$T_Q = \mathbf{r}_Q^T \mathbf{r}_Q^* T_Q^*,$$

$$\tau_Q = -\dot{\omega}_Q^* + \frac{k_3}{L_3} \mathbf{z}_\omega + \frac{L_2}{L_3} S(\mathbf{e}_3) \mathbf{R}_Q^T \mathbf{z}_Q,$$

where $\mathbf{r}_Q^*$ and T_Q^* are defined through (11), the origin of the error system is exponentially stable and the load position converges to the desired trajectory.

Proof. The closed-loop quadrotor and load error dynamic system, with states $\mathbf{z}_L$, $\mathbf{z}_Q$, and $\mathbf{z}_\omega$ defined as (10), (12), and (13), respectively, admits the Lyapunov function

$$V_3(\mathbf{z}_L, \mathbf{z}_Q, \mathbf{z}_\omega) = V(\mathbf{z}_L) + L_2 \frac{1}{2} \mathbf{z}_Q^T \mathbf{z}_Q + L_3 \frac{1}{2} \mathbf{z}_\omega^T \mathbf{z}_\omega \quad (14)$$

with the negative semi-definite time derivative

$$\dot{V}_3(\mathbf{z}_L, \mathbf{z}_Q, \mathbf{z}_\omega) \leq -W_3(\mathbf{z}_L, \mathbf{z}_Q, \mathbf{z}_\omega),$$

where

$$W_3(\mathbf{z}_L, \mathbf{z}_Q, \mathbf{z}_\omega) = W(\mathbf{z}_L) + k_Q \mathbf{z}_Q^T \Pi_{\mathbf{r}_Q} \mathbf{z}_Q + k_\omega \mathbf{z}_\omega^T \mathbf{z}_\omega$$

is a positive semi-definite function for $k_Q, k_\omega > 0$. Since $\dot{V}_3 \leq 0$, it follows that V_3 is bounded and so are all the error states. Adding the boundedness of the desired path $\mathbf{p}_d$ and its time derivatives we get that all quadrotor-load system states and their derivatives (1) are also bounded. Consequently, $\dot{V}_3$ is also bounded which implies that $\dot{V}_3$ is uniformly continuous. Applying Barbalat's Lemma it follows directly that V_3 , a uniformly continuous function, converges to zero as time flows. The closed-loop trajectories $(\mathbf{z}_L(t), \mathbf{z}_Q(t), \mathbf{z}_\omega(t))$ converge to one of the equilibrium points for which $W_3(\mathbf{z}) = 0$ yielding $\mathbf{p}_L(t) = \mathbf{p}_d(t)$, $\mathbf{r}_Q = \pm \mathbf{r}_Q^*$ and $\Pi_{\mathbf{e}_3} \omega_Q = \Pi_{\mathbf{e}_3} \omega_Q^*(t)$. Let c be the critical threshold for which $V(\mathbf{z}_L) < c$ implies $\dot{V}(\mathbf{z}_L)$ is strictly negative definite. Then, for initial conditions such that $V_3(\mathbf{z}_L, \mathbf{z}_Q, \mathbf{z}_\omega) < \min\{c, 2L_2\}$, the time derivative of V_3 is strictly negative definite, which implies that the origin is uniformly asymptotically stable.

4. Results

A simulation environment was developed in Matlab/Simulink to attest the performance of the proposed controller in ideal conditions and determine appropriate initial gains for the experimental tests. The vehicle and load system was modeled using the dynamics equations (1). Throughout the simulation the tension on the load T_L was monitored to ensure the string was taut

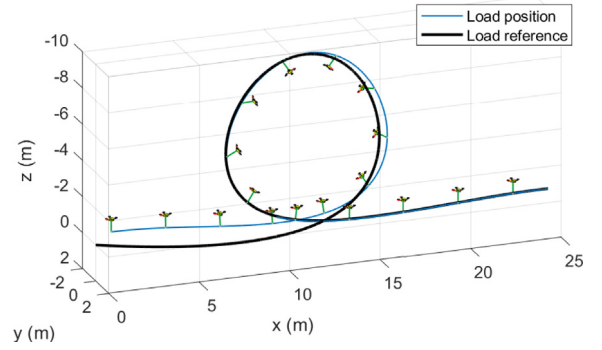


Fig. 2. Open-ended loop reference and simulation trajectories for the load with the quadrotor, string, and load position superimposed.

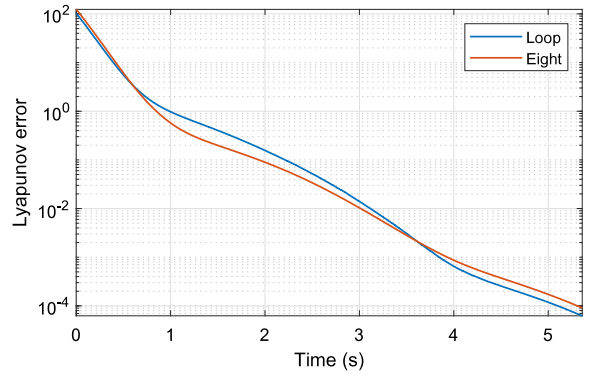


Fig. 3. Lyapunov function for the closed-curve and the loop-the-loop trajectories exhibiting exponential convergence to zero.

at all times and the dynamics were valid. The controller gains and Lyapunov parameters are the same for both simulations and are identical to the ones used for the experimental results. To increase the realism of the simulations the simulated quadrotor and load system parameters are identical to the real quadrotor and load system, $m_Q = 209$ g, $m_L = 47$ g, and $\ell = 0.70$ m, corresponding to a natural oscillation period of 1.7 s.

We present the results for two simulations, corresponding to two different trajectories to be tracked. The first trajectory is the closed curve in 3D space

$$\mathbf{p}_d(t) = [0.75 \sin(3t) \quad 1.5 \cos(1.5t) \quad \sin(1.5t)]^T.$$

The trajectory is performed in a confined space, has high curvature, and compels the load to move in all directions including up and down. The second is an aggressive open-ended trajectory, described by a quintic spline, where the load performs a *loop-the-loop* maneuver at a constant speed of 10 m s^{-1} , presented in Fig. 2. The tracking qualities of the proposed controller are evidenced by the exponential convergence of the Lyapunov function (14) depicted in Fig. 3. Despite the very different trajectories the convergence rate is similar, corresponding to a decrease of 6 orders of magnitude per 5 s.

The experimental results were obtained at the SCORE laboratory of the University of Macau with a model-scale radio controlled quadrotor controlled in thrust and angular velocity. Measurements were obtained using a motion capture setup, working at a 100 Hz rate, that provides accurate position, orientation, and velocities for the quadrotor vehicle and the slung load. These measurements are made available to a computer system which computes the quadrotor inputs according to the proposed control laws. The inputs (T_Q, ω_Q) are then transmitted

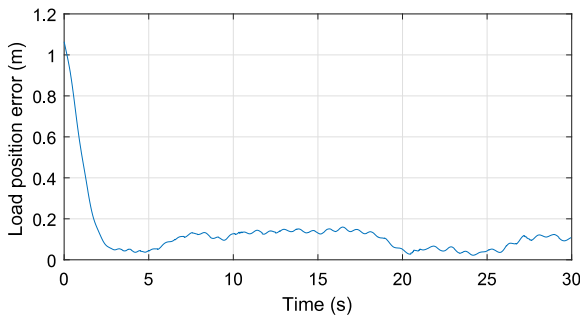


Fig. 4. Position error when tracking a figure eight trajectory. The initial transient is of short duration and the steady state error is small, averaging under 10 cm.

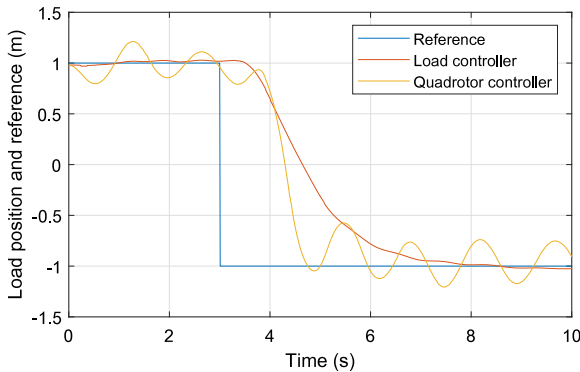


Fig. 5. Reference square wave trajectory highlighting the fast transient and the stability of the proposed controller and the subsiding undamped error when using a free flight quadrotor controller, ignoring the load effects.

to the vehicle at a 45 Hz rate through a radio control subsystem. The neglected input dynamics include a delay of the input response, similar to a low-pass filter with a pole around 0.1 s, and the dependence of T_Q with the battery voltage. The experiment also assumes zero communication and measurement acquisition delays. In practice, the communication and computing delays in the system are well within one sample time of the RC commands and can be neglected without performance detriment.

The reference trajectory is a figure eight

$$\mathbf{p}_d(t) = [0.75 \sin(0.6t) \quad 1.5 \cos(0.3t) \quad -0.9]$$

and the tracking error transient and steady-state can be seen in Fig. 4. The transient convergence to the path occurs within 2 s and the steady state error is around 10 cm. A comparison with a controller for the quadrotor position, emphasizing the improvements of using a controller for the load compared to a naive controller designed solely for the quadrotor, is presented in Fig. 5.

The aggressiveness of the tracking maneuver is tuned through the controller gains. We prioritized tracking speed in detriment of smoothness of the transient maneuver to introduce as little conservatism as possible, leading to fast maneuvers and high angles with the vertical for the string connecting the vehicle to the load. During the step maneuver the maximum angle with the vertical for $\mathbf{r}_l^*$ is 33° and for $\mathbf{r}_l$ is 24° .

5. Conclusions

We presented a trajectory tracking controller for the full dynamics of a quadrotor vehicle carrying a slung load. The quadrotor with slung load system is segmented in load, string, and

quadrotor subsystems. A parallel is drawn between the load and string subsystem and a free flying quadrotor that allows existing quadrotor trajectory tracking controllers to be used as a basis for the full controller. The actuation error introduced by the underactuation of the quadrotor is then backstepped through the quadrotor subsystem dynamics resulting in a closed-loop error system that admits a parameterized Lyapunov function with a negative semi-definite time derivative. The proposed controller is tunable, allowing the linear and angular errors to be appropriately mixed, drives the load position error to zero, and the origin of the error system is asymptotically stable. The controller was further implemented in simulation and experimental setups, attesting to its practical validity, performance and robustness. A comparison with a naive controller for a quadrotor only system was also presented where the performance differences are staggering.

Acknowledgments

This work was supported by the Macao Science and Technology Development Fund under Grant FDCT/026/2017/A1, by the University of Macau, Macao, China, under Project MYRG2018-00198-FST, by the Fundação para a Ciência e a Tecnologia (FCT) through ISR under Grant LARSyS UID/EEA/50009/2019, and FCT project LOTUS-PTDC/EEIAUT/5048/2014. The work of Rita Cunha was supported by the FCT Investigator Programme IF/00921/2013.

References

- Bernard, M., & Kondak, K. (2009). Generic slung load transportation system using small size helicopters. In *2009 IEEE international conference on robotics and automation* (pp. 3258–3264).
- Bisgaard, Morten, Bendtsen, Jan Dimon, & Cour-Harbo, Anders La (2009). Modeling of generic slung load system. *Journal of Guidance, Control, and Dynamics*, 32(2), 573–585. <http://dx.doi.org/10.2514/1.36539>.
- Cabecinhas, D., Cunha, R., & Silvestre, C. (2015). A globally stabilizing path following controller for rotorcraft with wind disturbance rejection. *IEEE Transactions on Control Systems Technology*, 23(2), 708–714.
- Faust, A., Palunko, I., Cruz, P., Fierro, R., & Tapia, L. (2013). Learning swing-free trajectories for UAVs with a suspended load. In *2013 IEEE international conference on robotics and automation* (pp. 4902–4909).
- Hua, Minh-Duc, Hamel, T., Morin, P., & Samson, C. (2009). A control approach for thrust-propelled underactuated vehicles and its application to VTOL drones. *IEEE Transactions on Automatic Control*, 54(8), 1837–1853. <http://dx.doi.org/10.1109/TAC.2009.2024569>.
- Michael, Nathan, Fink, Jonathan, & Kumar, Vijay (2011). Cooperative manipulation and transportation with aerial robots. *Autonomous Robots*, 30(1), 73–86. <http://dx.doi.org/10.1007/s10514-010-9205-0>.
- Palunko, I., Cruz, P., & Fierro, R. (2012). Agile load transportation: Safe and efficient load manipulation with aerial robots. *IEEE Robotics Automation Magazine*, 19(3), 69–79. <http://dx.doi.org/10.1109/MRA.2012.2205617>.
- Sreenath, K., & Kumar, V. (2013). Dynamics, control and planning for cooperative manipulation of payloads suspended by cables from multiple quadcopter robots. In *Robotics: Science and Systems*.
- Sreenath, K., Lee, T., & Kumar, V. (2013a). Geometric control and differential flatness of a quadrotor UAV with a cable-suspended load. In *52nd IEEE conference on decision and control* (pp. 2269–2274).
- Sreenath, Koushil, Lee, Taeyoung, & Kumar, Vijay (2013b). Geometric control and differential flatness of a quadrotor UAV with a cable-suspended load (pp. 2269–2274). *IEEE*, <http://dx.doi.org/10.1109/CDC.2013.6760219>.
- Sreenath, K., Michael, N., & Kumar, V. (2013a). Trajectory generation and control of a quadrotor with a cable-suspended load - A differentially-flat hybrid system. In *2013 IEEE international conference on robotics and automation* (pp. 4888–4895).
- Sreenath, K., Michael, N., & Kumar, V. (2013b). Trajectory generation and control of a quadrotor with a cable-suspended load - A differentially-flat hybrid system. In *2013 IEEE international conference on robotics and automation* (pp. 4888–4895).
- Tang, S., & Kumar, V. (2015). Mixed Integer Quadratic Program trajectory generation for a quadrotor with a cable-suspended payload. In *2015 IEEE international conference on robotics and automation* (pp. 2216–2222).



David Cabecinhas received the Licenciatura and Ph.D. degrees in Electrical and Computer Engineering from the Instituto Superior Técnico (IST), Lisbon, Portugal, in 2006 and 2014, respectively. He has been a Researcher with the Institute for Systems and Robotics, LarSyS, Lisbon, since 2007. He is currently a Post-Doctoral Fellow with the Faculty of Science and Technology, University of Macau, Macau, China. His current research interests include nonlinear control, sensor-based and vision-based control with applications to autonomous aerial and surface vehicles, and modeling and identification

of aerial and surface vehicles.



Rita Cunha received the Licenciatura degree in Information Systems and Computer Engineering and the Ph.D. degree in Electrical and Computer Engineering from the Instituto Superior Técnico (IST), Universidade de Lisboa, Portugal, in 1998 and 2007, respectively. She is currently an Assistant Researcher with the Institute for Systems and Robotics, LARSyS, Lisbon, and an Invited Assistant Professor with the Department of Electrical and Computer Engineering of IST. Her research interests include nonlinear dynamical systems and control, multi-agent systems,

cooperative control, and vision-based control with application to autonomous aerial vehicles.



Carlos Silvestre received the Licenciatura degree in Electrical Engineering from the Instituto Superior Técnico (IST) of Lisbon, Portugal, in 1987 and M.Sc. degree in Electrical Engineering and Ph.D. degree in Control Science from the same school in 1991 and 2000, respectively. In 2011, he received the Habilitation in Electrical Engineering and Computers also from IST. Since 2000, he is with the Department of Electrical Engineering of the Instituto Superior Técnico, where he is currently an Associate Professor of Control and Robotics on leave. Since 2015, he is a Professor of the

Department of Electrical and Computers

Engineering of the Faculty of Science and Technology of the University of Macau. Over the past years, he has conducted research on the subjects of navigation guidance and control of air and underwater robots. His research interests include linear and nonlinear control theory, coordinated control of multiple vehicles, gain scheduled control, integrated design of guidance and control systems, inertial navigation systems, and mission control and real time architectures for complex autonomous systems with applications to unmanned air and underwater vehicles.